TECHNICAL REPORT

Al – Faleej Tower

CLIVET VRF System — Equipment Schedule & Technical Data

Project	Al – Faleej Tower	**Document**	Technical Report
System	VRF (Variable Refrigerant Flow) — Air Conditioning	**Reference**	96/VRF – CLIVET / 26-T
Equipment	CLIVET VC MAX KUW VRF	**Power Supply**	380 – 415 V / 3 Ph
Scope covered	Outdoor units, indoor units, refrigerant branch joints and controllers		

1. System Summary

19 VRF SYSTEMS	**22** OUTDOOR UNITS	**141** INDOOR UNITS	**141** THERMOSTATS
235.65 TR TOTAL ACTUAL CAPACITY (ODU) @ 48°C	**281.05 TR** TOTAL NOMINAL CAPACITY (IDU)	**4** SYSTEM ZONES	**125** BRANCH JOINTS

SYSTEM ZONE	DESIGNATION	ODU	IDU	THERMOSTATS
Basement	SYS.B-01	2	10	10
Ground Floor	SYS.GF-01	2	11	11
Mezzanine Floor	SYS.M-01	2	8	8
1st – 16th Floor (16 typical floors)	SYS.TY-01 to 16	16	112	112
Total		**22**	**141**	**141**

2. Equipment Schedule — SYS.B-01 | Basement

MODEL	QTY	UNIT TYPE	DESCRIPTION
CRVCM-XMi 500T (KUW)	1	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)
CRVCM-XMi 450T (KUW)	1	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)
CHT1-3-XYMi D400	1	Indoor (IDU)	High Static Pressure Duct
CHT2-3-XYMi D112	1	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D90	2	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D71	4	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D56	1	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D36	1	Indoor (IDU)	Medium Static Pressure Duct
FQZHW-02N1E	1	Branch Joint	Copper Y Joint
FQZHN-01D	2	Branch Joint	Copper Y Joint
FQZHN-02D	2	Branch Joint	Copper Y Joint
FQZHN-03D	5	Branch Joint	Copper Y Joint
WDC3-86S	10	Thermostat	3rd Generation Group Controller

3. Equipment Schedule — SYS.GF-01 | Ground Floor

MODEL	QTY	UNIT TYPE	DESCRIPTION
CRVCM-XMi 335T (KUW)	2	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)
CHT2-3-XYMi D90	3	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D71	6	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D56	1	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D28	1	Indoor (IDU)	Medium Static Pressure Duct
FQZHW-02N1E	1	Branch Joint	Copper Y Joint
FQZHN-01D	2	Branch Joint	Copper Y Joint
FQZHN-02D	1	Branch Joint	Copper Y Joint
FQZHN-03D	7	Branch Joint	Copper Y Joint
WDC3-86S	11	Thermostat	3rd Generation Group Controller

4. Equipment Schedule — SYS.M-01 | Mezzanine Floor

MODEL	QTY	UNIT TYPE	DESCRIPTION
CRVCM-XMi 335T (KUW)	2	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)
CHT2-3-XYMi D112	4	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D90	2	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D56	2	Indoor (IDU)	Medium Static Pressure Duct
FQZHW-02N1E	1	Branch Joint	Copper Y Joint
FQZHN-01D	1	Branch Joint	Copper Y Joint
FQZHN-02D	1	Branch Joint	Copper Y Joint
FQZHN-03D	5	Branch Joint	Copper Y Joint
WDC3-86S	8	Thermostat	3rd Generation Group Controller

5. Equipment Schedule — SYS.TY-01 to 16 | 1st Floor to 16th Floor

MODEL	QTY / FLOOR	TOTAL (16 FL.)	UNIT TYPE	DESCRIPTION
CRVCM-XMi 400T (KUW)	1	16	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)
CHT2-3-XYMi D112	1	16	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D90	1	16	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D71	2	32	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D56	1	16	Indoor (IDU)	Medium Static Pressure Duct
CHT2-3-XYMi D28	2	32	Indoor (IDU)	Medium Static Pressure Duct
FQZHN-01D	1	16	Branch Joint	Copper Y Joint
FQZHN-02D	5	80	Branch Joint	Copper Y Joint
WDC3-86S	7	112	Thermostat	3rd Generation Group Controller

Each typical floor (1st to 16th) is served by one independent VRF system of identical configuration.

6. Consolidated Equipment Schedule — Whole Project

MODEL	UNIT TYPE	DESCRIPTION	TOTAL QTY
CRVCM-XMi 500T (KUW)	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)	1
CRVCM-XMi 450T (KUW)	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)	1
CRVCM-XMi 400T (KUW)	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)	16
CRVCM-XMi 335T (KUW)	Outdoor (ODU)	VC MAX KUW VRF (380–415 V)	4
CHT1-3-XYMi D400	Indoor (IDU)	High Static Pressure Duct	1
CHT2-3-XYMi D112	Indoor (IDU)	Medium Static Pressure Duct	21
CHT2-3-XYMi D90	Indoor (IDU)	Medium Static Pressure Duct	23
CHT2-3-XYMi D71	Indoor (IDU)	Medium Static Pressure Duct	42
CHT2-3-XYMi D56	Indoor (IDU)	Medium Static Pressure Duct	20
CHT2-3-XYMi D36	Indoor (IDU)	Medium Static Pressure Duct	1
CHT2-3-XYMi D28	Indoor (IDU)	Medium Static Pressure Duct	33
FQZHW-02N1E	Branch Joint	Copper Y Joint	3
FQZHN-01D	Branch Joint	Copper Y Joint	21
FQZHN-02D	Branch Joint	Copper Y Joint	84
FQZHN-03D	Branch Joint	Copper Y Joint	17
WDC3-86S	Thermostat	3rd Generation Group Controller	141
Total Outdoor Units (ODU)			**22**
Total Indoor Units (IDU)			**141**
Total Branch Joints			**125**

7. Technical Notes

- The air conditioning system is a **CLIVET VC MAX KUW VRF** system, operating on a **380 – 415 V / 3 Phase** power supply.
- The building is divided into **4 system zones** — Basement, Ground Floor, Mezzanine Floor, and 16 typical floors — forming a total of **19 independent VRF systems**.
- Total actual outdoor unit capacity at an ambient design temperature of **48°C** is **235.65 TR**.
- Total nominal indoor unit capacity is **281.05 TR**, giving an indoor-to-outdoor capacity (diversity) ratio of approximately **119%**.
- Indoor units are of the **ducted type**: one high static pressure unit (D400) and 140 medium static pressure units.
- Each indoor unit is provided with one **WDC3-86S** 3rd generation group controller (thermostat), giving a 1:1 ratio across all 141 indoor units.
- Refrigerant distribution uses **copper Y branch joints** of the FQZHW / FQZHN series, as scheduled per system.
- This schedule covers outdoor units, indoor units, branch joints and controllers only. Interconnecting refrigerant copper piping, installation works and spare parts are not part of this equipment schedule.
- Quantities and capacities are based on the design data available at the date of this report. Any change in project conditions, served areas, or unit distribution requires a revision of this report.